

More Identities (Homework)

Name
Date
Course

Prove that the following equations are true by converting the left side to the right.

$$\textcircled{1} 1 + \cot^2(-A) = \csc^2 A \quad \textcircled{2} \frac{\csc(-\theta)}{\sin \theta} = -\csc^2 \theta$$

$$\textcircled{3} -2\tan x - \tan x = 3\tan(-x) \quad \textcircled{4} \frac{\sin w}{\cos w} \cdot \frac{\sin(-w)}{\cos(-w)} = -\tan^2 w$$

$$\textcircled{5} \cos(B-360^\circ)\cos(B-180^\circ) = -\cos^2 B \quad \textcircled{6} \frac{\sin(\theta+360^\circ)}{\sin(\theta-180^\circ)} = -1$$

$$\textcircled{7} \frac{\csc^2 x}{\csc(x+180^\circ) + \csc(x-180^\circ)} = -\frac{\csc x}{2} \quad \textcircled{8} \cot(\theta+360^\circ) - \cot \theta = 0$$

$$\textcircled{9} \sin^2(360^\circ - W) + \cos^2(180^\circ - W) = 1 \quad \textcircled{10} \sec(180^\circ - \theta) - 10 \sec \theta = -11 \sec \theta$$

$$\textcircled{11} \frac{\csc(360^\circ - x)}{\csc^2 x} = -\sin x \quad \textcircled{12} -\frac{\cos A}{\sin A} = \cot(180^\circ - A)$$

$$\textcircled{13} \tan(90^\circ - B) - 5 \cot B = -4 \cot B \quad \textcircled{14} \csc(\theta + 90^\circ) - \csc(\theta - 90^\circ) = 2 \sec \theta$$

$$\textcircled{15} \cot^2(90^\circ - x) + \sin^2 x + \cos^2 x = \sec^2 x \quad \textcircled{16} \frac{1}{\cos(A - 90^\circ)} + \frac{1}{\cos(A + 90^\circ)} = 0$$

Use the sum and difference formulas to derive the following identities.

$$(17) \sin(180^\circ - x) = \sin x$$

$$\sin(A - B) = \sin A \cos B - \sin B \cos A$$

$$(18) \sec(\theta - 90^\circ) = \csc \theta$$

$$\sec(A - B) = \frac{1}{\cos(A - B)} = \frac{1}{\cos A \cos B + \sin A \sin B}$$

$$(19) \tan(W + 90^\circ) = -\cot W$$

$$\tan(A + B) = \frac{\sin(A + B)}{\cos(A + B)} = \frac{\sin A \cos B + \sin B \cos A}{\cos A \cos B - \sin A \sin B}$$

(20) Prove the fourth product to sum formula by using the following equation.

$$\cos(A - B) - \cos(A + B) = \cos A \cos B + \sin A \sin B - [\cos A \cos B - \sin A \sin B]$$

$$\frac{1}{2}[\cos(A - B) - \cos(A + B)] = \sin A \sin B \leftarrow \text{Destination.}$$

Homework Answers

Prove that the following equations are true by converting the left side to the right.

$$\textcircled{1} 1 + \cot^2(-A) = \csc^2 A \quad \textcircled{2} \frac{\csc(-\theta)}{\sin \theta} = -\csc^2 \theta$$

$$1 + (\cot(-A))^2 =$$

$$1 + (-\cot A)^2 =$$

$$1 + \cot^2 A =$$

$$\csc^2 A =$$



$$\frac{-\csc \theta}{\sin \theta} =$$

$$-\csc \theta \div \sin \theta =$$

$$-\frac{1}{\sin \theta} \div \sin \theta =$$

$$-\frac{1}{\sin \theta} \cdot \frac{1}{\sin \theta} =$$

$$-\frac{1}{\sin^2 \theta} =$$

$$-\csc^2 \theta =$$



$$\textcircled{3} -2\tan x - \tan x = 3\tan(-x) \quad \textcircled{4} \frac{\sin w}{\cos w} \cdot \frac{\sin(-w)}{\cos(-w)} = -\tan^2 w$$

$$2(-\tan x) - \tan x =$$

$$2\tan(-x) + \tan(-x) =$$

$$3\tan(-x) =$$



$$\tan w \cdot \frac{-\sin w}{\cos w} =$$

$$\tan w \cdot (-\tan w) =$$

$$-\tan^2 w =$$



$$\textcircled{5} \cos(B-360^\circ)\cos(B-180^\circ) = -\cos^2 B \quad \textcircled{6} \frac{\sin(\theta+360^\circ)}{\sin(\theta-180^\circ)} = -1$$

$$\cos B (-\cos B) =$$

$$-\cos^2 B =$$



$$\frac{\sin \theta}{-\sin \theta} =$$

$$-1 =$$



$$\textcircled{7} \frac{\csc^2 x}{\csc(x+180^\circ) + \csc(x-180^\circ)} = -\frac{\csc x}{2} \quad \textcircled{8} \cot(\theta+360^\circ) - \cot \theta = 0$$

$$\frac{\csc^2 x}{-\csc x - \csc x} =$$

$$\frac{\csc^2 x}{-2\csc x} =$$

$$-\frac{\csc x}{2} =$$



$$\cot \theta - \cot \theta =$$

$$0 =$$



$$(9) \sin^2(360^\circ - w) + \cos^2(180^\circ - w) = 1 \quad (10) \sec(180^\circ - \theta) - 10\sec\theta = -11\sec\theta$$

$$(\sin(360^\circ - w))^2 + (\cos(180^\circ - w))^2 = -\sec\theta - 10\sec\theta =$$

$$(-\sin w)^2 + (-\cos w)^2 = -11\sec\theta =$$

$$\sin^2 w + \cos^2 w =$$

$$1 =$$

$$(11) \frac{\csc(360^\circ - x)}{\csc^2 x} = -\sin x \quad (12) -\frac{\cos A}{\sin A} = \cot(180^\circ - A)$$

$$\frac{-\csc x}{\csc^2 x} =$$

$$-\cot A =$$

$$-\frac{1}{\csc x} =$$

$$\cot(180^\circ - A) =$$

$$-\sin x =$$

$$(13) \tan(90^\circ - B) - 5\cot B = -4\cot B \quad (14) \csc(\theta + 90^\circ) - \csc(\theta - 90^\circ) = 2\sec\theta$$

$$\cot B - 5\cot B =$$

$$\sec\theta - (-\sec\theta) =$$

$$-4\cot B =$$

$$\sec\theta + \sec\theta =$$

$$2\sec\theta =$$

$$(15) \cot^2(90^\circ - x) + \sin^2 x + \cos^2 x = \sec^2 x \quad (16) \frac{1}{\cos(A - 90^\circ)} + \frac{1}{\cos(A + 90^\circ)} = 0$$

$$\tan^2 x + 1 =$$

$$\frac{1}{\sin A} - \frac{1}{\sin A} =$$

$$\sec^2 x =$$

$$0 =$$

Use the sum and difference formulas to derive the following identities.

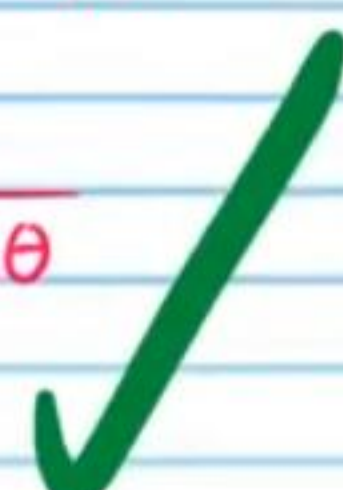
$$(17) \sin(180^\circ - x) = \sin x$$

$$\sin(A-B) = \sin A \cos B - \sin B \cos A$$

$$\begin{aligned}\sin(180^\circ - x) &= \sin 180^\circ \cos x - \sin x \cos 180^\circ \\ &= (0) \cos x - (\sin x)(-1) \\ &= 0 + \sin x \\ &= \sin x\end{aligned}$$


$$(18) \sec(\theta - 90^\circ) = \csc \theta$$

$$\sec(A-B) = \frac{1}{\cos(A-B)} = \frac{1}{\cos A \cos B + \sin A \sin B}$$

$$\begin{aligned}\sec(\theta - 90^\circ) &= \frac{1}{\cos(\theta - 90^\circ)} = \frac{1}{\cos \theta \cos 90^\circ + \sin \theta \sin 90^\circ} \\ &= \frac{1}{(\cos \theta)(0) + (\sin \theta)(1)} \\ &= \frac{1}{0 + \sin \theta} \\ &= \csc \theta\end{aligned}$$


$$(19) \tan(w + 90^\circ) = -\cot w$$

$$\tan(A+B) = \frac{\sin(A+B)}{\cos(A+B)} = \frac{\sin A \cos B + \sin B \cos A}{\cos A \cos B - \sin A \sin B}$$

$$\begin{aligned}\tan(w + 90^\circ) &= \frac{\sin(w + 90^\circ)}{\cos(w + 90^\circ)} = \frac{\sin w \cos 90^\circ + \sin 90^\circ \cos w}{\cos w \cos 90^\circ - \sin w \sin 90^\circ} \\ &= \frac{(\sin w)(0) + (1) \cos w}{(\cos w)(0) - (\sin w)(1)} \\ &= \frac{0 + \cos w}{0 - \sin w} \\ &= -\frac{\cos w}{\sin w} \\ &= -\cot w\end{aligned}$$


(20)

$$\cos(A-B) - \cos(A+B) = \cos A \cos B + \sin A \sin B - [\cos A \cos B - \sin A \sin B]$$

$$\begin{aligned}\cos(A-B) - \cos(A+B) &= \cancel{\cos A \cos B} + \sin A \sin B - \cancel{\cos A \cos B} + \sin A \sin B \\ &= \sin A \sin B + \sin A \sin B\end{aligned}$$

$$\cos(A-B) - \cos(A+B) = 2 \sin A \sin B$$

$$\frac{1}{2} [\cos(A-B) - \cos(A+B)] = \sin A \sin B$$

